1-to'plam. *Nuqta qaysi chorakka tegishli ekanini aniqlang.*

1. $(3,\ 5)$ → chorak:

2. $(-2,\ 4)$ → chorak:

3. $(-6,\ -1)$ → chorak:

4. $(7,\ -3)$ → chorak:

5. $(-4,\ -8)$ → chorak:

6. $(5,\ -5)$ → chorak:

7. $(-9,\ 2)$ → chorak:

8. $(1,\ 1)$ → chorak:

2-to'plam. *Tenglamadan k (burchak koeffitsienti) va b ni ajrating.*

1. $y = 3x + 7$ → $k=$ $b=$

2. $y = -2x + 1$ → $k=$ $b=$

3. $y = x - 5$ → $k=$ $b=$

4. $y = -x + 9$ → $k=$ $b=$

5. $y = 4x$ → $k=$ $b=$

6. $y = -6$ → $k=$ $b=$

7. $y = 0.5x - 3$ → $k=$ $b=$

8. $y = \frac{1}{3}x - 2$ → $k=$ $b=$

3-to'plam. *$f(a)$ ni musbat qiymatlar uchun hisoblang.*

1. $f(x) = 2x + 3.\ f(4) =$

2. $f(x) = 5x - 1.\ f(2) =$

3. $f(x) = -3x + 4.\ f(1) =$

4. $f(x) = x + 10.\ f(0) =$

5. $f(x) = 4x.\ f(3) =$

6. $f(x) = -x + 6.\ f(5) =$

7. $f(x) = 2x - 7.\ f(10) =$

8. $f(x) = -5x + 2.\ f(3) =$

4-to'plam. *$f(a)$ ni manfiy qiymatlar uchun hisoblang – manfiy son har doim qavsda.*

1. $f(x) = 3x + 2.\ f(-2) =$

2. $f(x) = -4x + 1.\ f(-3) =$

3. $f(x) = x + 8.\ f(-8) =$

4. $f(x) = x^2.\ f(-3) =$

5. $f(x) = x^2 + 1.\ f(-4) =$

6. $f(x) = 2x^2 - 3.\ f(-2) =$

7. $f(x) = -x^2.\ f(-5) =$

8. $f(x) = (x + 1)^2.\ f(-6) =$

5-to'plam. *Aralash – shaklni almashtiring, qiymatni hisoblang, tenglamani yeching.*

1 $y = 5x - 3$ ni $f(x)$ shaklida yozing va $f(2)$ ni toping.

2 $f(x) = -2x + 8$ ni $y = kx + b$ shaklida yozing, k, b ni toping.

3 $y = x + 1$ ni $f(x)$ shaklida yozing, so'ng $f(-4)$ ni hisoblang.

4 $f(x) = 4x - 9$. Agar $f(x) = 11$ bo'lsa, x ni toping.

5 $f(x) = -3x + 6$. Agar $f(x) = 0$ bo'lsa, x ni toping.

6 $y = -x - 5$ ni $f(x)$ shaklida yozing va $f(0)$ ni toping.

7 $f(x) = 2x + 1$ bo'lsa, $f(3) + f(-3)$ ni hisoblang.

8 $f(x) = x^2 - 4$. Agar $f(a) = 5$ bo'lsa, a ning musbat qiymatini toping.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $f(x) = 3x + 2$. $f(5)$ ni toping:

"$f(5) = f \cdot 5$, f noma'lum, javob topib bo'lmaydi."

Xato va to'g'ri javob: .....

2 $f(x) = x^2$. $f(-4)$ ni toping:

"$f(-4) = -4^2 = -16$."

Xato va to'g'ri javob: .....

3 $(-3,\ 5)$ nuqtasi qaysi chorakda?

"$x = -3$ ni musbat deb o'ylab, I chorak."

Xato va to'g'ri javob: .....

4 $f(x) = 2x - 5$. Agar $f(x) = 7$, x ni toping:

"$2x - 5 = 7 \Rightarrow 2x = 2 \Rightarrow x = 1$."

Xato va to'g'ri javob: .....

5 $f(x) = x^2 + 3$. $f(-2)$ ni toping:

"$f(-2) = -2^2 + 3 = -4 + 3 = -1$."

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1 $f(x) = 2x^2 - 3x + 1$. $f(-2) =$

2 $f(x) = -(x-3)^2$. $f(6) =$

3 $y = -4x + 11$ ni $f(x)$ shaklida yozing, so'ng $f(x) = 3$ bo'lganda x ni toping.

4 $f(x) = x^2 - 2x$. Agar $f(a) = 8$ bo'lsa, a ning manfiy qiymatini toping.

5 $f(x) = 3x + c$. Agar $f(2) = 10$ bo'lsa, c ni toping.

8-to'plam. *SAT'da qanday uchraydi – chorak, qiyalik va funksiya qiymati.*

1 $(-5,\ 3)$ nuqtasi qaysi chorakda joylashgan?

(A) II (B) I

(C) III (D) IV

2 $y = -2x + 9$ to'g'ri chizig'ining qiyaligi (k) nechaga teng?

(A) 2 (B) -2

(C) 9 (D) -9

3 $y = 7x - 4$ to'g'ri chizig'ining erkin hadi (b) nechaga teng?

(A) 7 (B) 4

(C) -4 (D) -7

4 Agar $f(x) = 3x - 5$ bo'lsa, $f(4)$ nechaga teng?

(A) 12 (B) 17

(C) -5 (D) 7

5 Agar $f(x) = x^2 + 2$ bo'lsa, $f(-3)$ nechaga teng?

(A) 11 (B) -7

(C) 7 (D) -11

6 $(0,\ -6)$ nuqtadan o'tuvchi va qiyaligi 4 ga teng bo'lgan chiziq tenglamasi qaysi?

(A) $y=-6x+4$ (B) $y=4x-6$

(C) $y=4x+6$ (D) $y=-4x-6$

7 $(6,\ -2)$ nuqtasi qaysi chorakda joylashgan?

(A) I (B) II

(C) IV (D) III

8 $y = -x + 3$ to'g'ri chizig'i Oy o'qini qaysi nuqtada kesib o'tadi?

(A) $(3, 0)$ (B) $(0, -1)$

(C) $(-1, 0)$ (D) $(0, 3)$

9 Agar $g(x) = 2x + 1$ bo'lsa, $g(5) - g(1)$ nechaga teng?

(A) 8 (B) 4

(C) 14 (D) 2

10 $y = \frac{1}{2}x - 4$ to'g'ri chizig'ining qiyaligi (k) nechaga teng?

(A) -4 (B) $\frac{1}{2}$

(C) 2 (D) $-\frac{1}{2}$

11 Agar $h(x) = -x^2 + 4$ bo'lsa, $h(-2)$ nechaga teng?

(A) 8 (B) -4

(C) 0 (D) 4

12 $y = 3x - 2$ to'g'ri chizig'ida qaysi nuqta yotadi?

(A) $(2, 6)$ (B) $(4, 2)$

(C) $(2, -4)$ (D) $(2, 4)$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	I	2	II
	3	III	4	IV
	5	III	6	IV
	7	II	8	I
2-to'plam	1	k=3, b=7	2	k=−2, b=1
	3	k=1, b=−5	4	k=−1, b=9
	5	k=4, b=0	6	k=0, b=−6
	7	k=0.5, b=−3	8	$k=\frac{1}{3}$, b=−2
3-to'plam	1	11	2	9
	3	1	4	10
	5	12	6	1
	7	13	8	−13
4-to'plam	1	−4	2	13
	3	0	4	9
	5	17	6	5
	7	−25	8	25
5-to'plam	1. $f(x) = 5x-3$; $f(2) = 7$ 2. $y = -2x+8$; $k = -2$, $b = 8$			
	3. $f(x) = x + 1$; $f(-4) = -3$ 4. $x = 5$			
	5. $x = 2$ 6. $f(x) = -x - 5$; $f(0) = -5$			
	7. $f(3) + f(-3) = 7 + (-5) = 2$			
	8. $a = 3$			
7-to'plam	1	15	2	−9
	3. $f(x) = -4x + 11$; $x = 2$ 4. $a = -2$ 5. $c = 4$			
8-to'plam	1	A	2	B
	3	C	4	D
	5	A	6	B
	7	C	8	D
	9	A	10	B
	11	C	12	D
6-to'plam	1. f funksiya nomi, songa ko'paytirilmaydi. To'g'ri: $f(5) = 3(5) + 2 = 17$.			
	2. Manfiy kiritmada qavs tushib qolgan. To'g'ri: $f(-4) = (-4)^2 = 16$.			
	3. x ning ishorasi noto'g'ri o'qilgan. To'g'ri: $x = -3<0$, $y=5>0 \Rightarrow$ II chorak.			
	4. $7 + 5 = 12$, 2 emas. To'g'ri: $2x = 12 \Rightarrow x = 6$.			
	5. Yana qavs tushib qolgan. To'g'ri: $f(-2) = (-2)^2+3 = 7$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

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Telegram